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AI-Rad Companion Brain MR VA40x

D&A

Summary Technical Documentation (STED)

Regulatory Document

R1.0

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1 Introduction

1.1 Purpose

The product in scope of this document is a medical device within the meaning of Art. 2 (1) of Regulation (EU) 2017/745 on Medical Devices (EU MDR).

The purpose of this document is to confirm and document the conformity assessment with regards to EU MDR as well as other applicable European legislation, and to provide the related technical documentation needed to meet defined requirements.

1.2 Scope

This document is valid for AI-Rad Companion Brain MR, Version VA40x[1].

Due to the nature of D&A Medical Devices (all are considered as "Software as a Medical Device"), the RoHS directive as well as other Hardware related standards and regulation requirements are not applicable.

Project documents are stored in the internal document management system eDoc. The full list of documents and their revisions are available in the corresponding Document Plan [4]. Process documents are accessible through the QMS intranet pages.

Note: AI-Rad Companion Brain MR is a software-only medical device which is developed under the umbrella of AI-Rad Companion product family. Hence, while some documents are valid specifically for this device application, some documents may be valid for several devices part of the family.

1.3 Definitions, Abbreviations and Acronyms

1.3.1 Definitions and Acronyms

Term	Definition
Accessory for a medical device	'accessory for a medical device' means an article which, whilst not being itself a medical device, is intended by its manufacturer to be used together with one or several particular medical device(s) to specifically enable the medical device(s) to be used in accordance with its/their intended purpose(s) or to specifically and directly assist the medical functionality of the medical device(s) in terms of its/their intended purpose(s).
Active device	'active device' means any device, the operation of which depends on a source of energy other than that generated by the human body for that purpose, or by gravity, and which acts by changing the density of or converting that energy. Devices intended to transmit energy, substances or other elements between an active device and the patient, without any significant change, shall not be deemed to be active

[1] Where "x" indicates bug-fixing versions only, means, defects solved to restore device to original specifications, therefore do not trigger an update of this STED. Defects solved will in general be traceable and documented in a scope of update document and corresponding deliverables in separate specific Document Plan.

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Term	Definition
	devices.
AI	Artificial Intelligence
Authorized representative	‘authorized representative’ means any natural or legal person established within the Union who has received and accepted a written mandate from a manufacturer, located outside the Union, to act on the manufacturer's behalf in relation to specified tasks with regard to the latter's obligations under the EU MDR.
Clinical data	‘clinical data’ means information concerning safety or performance that is generated from the use of a device and is sourced from the following: – clinical investigation(s) of the device concerned, – clinical investigation(s) or other studies reported in scientific literature, of a device for which equivalence to the device in question can be demonstrated, – reports published in peer reviewed scientific literature on other clinical experience of either the device in question or a device for which equivalence to the device in question can be demonstrated, – clinically relevant information coming from post-market surveillance, in particular the post-market clinical follow-up.
Clinical evaluation	‘clinical evaluation’ means a systematic and planned process to continuously generate, collect, analyze and assess the clinical data pertaining to a device in order to verify the safety and performance, including clinical benefits, of the device when used as intended by the manufacturer.
Common Specifications	‘common specifications’ (CS) means a set of technical and/or clinical requirements, other than a standard, that provides a means of complying with the legal obligations applicable to a device, process or system.
Configurable devices	A configurable device is a device that consists of several components which can be assembled by the manufacturer in multiple configurations. Those individual components may be devices in themselves. Configurable devices include computed tomography (CT) systems, ultrasound systems, anesthesia systems, physiological Monitoring systems, radiology information systems (RIS).
Configuration	Configuration is a combination of items of equipment, as specified by the manufacturer, that operate together as a device to achieve an intended purpose. The combination of items may be modified, adjusted or customized to meet specific needs. Configurations include *inter alia*: – gantries, tubes, tables, consoles and other items of equipment that can be configured/combined to deliver an intended function in computed tomography.

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Term	Definition
	– ventilators, breathing circuits, vaporizers combined to deliver an intended function in anesthesia.
Conformity assessment	‘conformity assessment’ means the process demonstrating whether the requirements of EU MDR relating to a device have been fulfilled.
Economic Operator	‘economic operator’ means a manufacturer, an authorized representative, an importer, a distributor or the person referred to in Article 22(1) and 22(3) EU MDR.
EMDN-Code/-Term	The “European Medical Device Nomenclature (EMDN)” is a European product coding system for medical devices and IVDs under EU MDR/IVDR based on the Italian system CLASSIFICAZIONE NAZIONALE DISPOSITIVI MEDICI (called CND).
EU Declaration of conformity	A continuously updated declaration by the manufacturer that shall state that the requirements specified in EU MDR have been fulfilled in relation to the device that is covered.
EUDAMED	European database on medical devices, that include the following electronic systems: for registration of devices; for UDI-database; for registration of economic operators, for notified bodies and for certificates, for clinical investigations, for vigilance and post-market surveillance, for market surveillance.
Facility(ies)	internal organization/location that is responsible for the manufacturing of the device acc. to the EC Certificate of the legal manufacturer.
GMDN-Code	The “Global Medical Device Nomenclature (GMDN)” is a comprehensive system of internationally recognized coded descriptors in the format of preferred terms with definitions used to generically identify medical devices and related health care products.
Harmonized standards	‘harmonized standard’ means a European standard as defined in point (1)(c) of Article 2 of Regulation (EU) No 1025/2012.
Importer	‘importer’ means any natural or legal person established within the Union that places a device from a third country on the Union market.
Instructions for use	‘instructions for use’ means the information provided by the manufacturer to inform the user of a device's intended purpose and proper use and of any precautions to be taken.
Intended purpose	‘intended purpose’ means the use for which a device is intended according to the data supplied by the manufacturer on the label, in the instructions for use or in promotional or sales materials or statements and as specified by the manufacturer in the clinical evaluation. *Note:* For the content of the intended purpose also Annex I, chapter III, 23.4 needs to be considered.
Label	‘label’ means the written, printed or graphic information appearing either on the device itself, or on the packaging of each unit or on the packaging of multiple devices.
Manufacturer	‘manufacturer’ means a natural or legal person who manufactures or

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Term	Definition
	fully refurbishes a device or has a device designed, manufactured or fully refurbished, and markets that device under its name or trademark.
Medical Device	'medical device' means any instrument, apparatus, appliance, software, implant, reagent, material or other article intended by the manufacturer to be used, alone or in combination, for human beings for one or more of the following specific medical purposes: – diagnosis, prevention, monitoring, prediction, prognosis, treatment or alleviation of disease, – diagnosis, monitoring, treatment, alleviation of, or compensation for, an injury or disability, – investigation, replacement or modification of the anatomy or of a physiological or pathological process or state, – providing information by means of in vitro examination of specimens derived from the human body, including organ, blood and tissue donations, and which does not achieve its principal intended action by pharmacological, immunological or metabolic means, in or on the human body, but which may be assisted in its function by such means. The following products shall also be deemed to be medical devices: – devices for the control or support of conception; – products specifically intended for the cleaning, disinfection or sterilization of devices as referred to in Article 1(4) and of those referred to in the first paragraph of this point.
Medical Device Software (MDSW)	Medical device software is software that is intended to be used, alone or in combination, for a purpose as specified in the definition of a "medical device" in the medical devices regulation or in vitro diagnostic medical devices regulation.
MDR Annex XVI device	Annex XVI of the EU MDR regulation (EU MDR defined in [1]) lists out the groups of products without an intended medical purpose, which are also covered under the scope of the EU MDR.
National Competent Authorities	A country's regulatory body (e.g., the MHRA in the UK, BfArM in Germany, etc.) that is charged with monitoring compliance with the national statutes and regulations of European Member States.
Notified Body	'notified body' means a conformity assessment body designated in accordance with EU MDR.
Post market surveillance	'post-market surveillance' means all activities carried out by manufacturers in cooperation with other economic operators to institute and keep up to date a systematic procedure to proactively collect and review experience gained from devices they place on the market, make available on the market or put into service for the purpose of identifying any need to immediately apply any necessary corrective or preventive actions.
Single Registration	A Single Registration Number (SRN) is a Number to identify the manufacturer and where applicable the EU AR and the importer,

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Term	Definition
Number	provided by the EUDAMED.
Unique Device Identifier	'Unique Device Identifier' ('UDI') means a series of numeric or alphanumeric characters that is created through internationally accepted device identification and coding standards and that allows unambiguous identification of specific devices on the market.

For further definitions see DH Glossary.

1.3.2 Abbreviations

Abbreviation	Definition
AIRC	AI-Rad Companion
CAP	Conformity Assessment Procedure
CS	Common Specifications
CSF	Cerebro Spinal Fluid
CUT	Customer Use Test
D&A	Digital & Automation
DICOM	Digital Imaging and Communication in Medicine
DMR	Device Master Record
EU DoC	Declaration of Conformity
EMDN	European Medical Device Nomenclature
EU MDR	Regulation (EU) 2017/745 of the European Parliament and the council of 4 April 2017
EUDAMED	European database on medical devices
GMDN	Global Medical Device Nomenclature
GSPR	General Safety and Performance Requirements
GTIN	Global Trade Item Number
IEC	International Electrotechnical Commission
IFU	Instruction(s) for Use
MDSW	Medical Device Software
MPRAGE	Magnetization Prepared Rapid Acquisition Gradient Echo
MR	Magnetic Resonance
MRI	Magnetic Resonance Imaging
n.a.	Not applicable
PACS	Picture Archiving and Communication System
PLM	Product life cycle management
PMCF	Post Market Clinical Follow-up

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Abbreviation	Definition
PMS	Post Market Surveillance
PRRC	Person Responsible for Regulatory Compliance
PSS	Product and Solution Security
PSUR	periodic safety update reports
RIS	Radiology Information System
RoHS	Directive on the restriction of the use of certain hazardous substances in electrical and electronic equipment (EC Council Directive 2011/65/EU of July 01, 2011)
SRN	Single Registration Number
STED	Summary Technical Documentation
UDI UDI-DI UDI-PI	Unique Device Identification Unique Device Identification – Device Identifier Unique Device Identification – Production Identifier

For further definitions see D&A QMS Glossary.

1.4 References

1.4.1 Project Documents

Refer to Section 13.

1.4.2 Process Documents

Refer to Section 13.17

1.4.3 Further references – informative

[1] Medical Device Regulation (EU) 2017/745 (MDR)

[2] MDCG 2019-11 - Guidance on Qualification and Classification of Software in Regulation (EU) 2017/745 - MDR and Regulation (EU) 2017/746, October 2019.

[3] Commission Regulation (EU) No. 2021/2226 of 14 December 2021 laying down rules for the application of Regulation (EU) 2017/745 of the European Parliament and of the Council as regards electronic instructions for use of medical devices

1.5 Transitional Arrangements

n.a.

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2 Device Description and Specification

2.1 Product or Trade name

AI-Rad Companion Brain MR

2.1.1 Intended Purpose

2.1.1.1 Intended Purpose statement according to EC certificate

Not applicable. Device is a class IIa device.

2.1.1.2 Intended Use

AI-Rad Companion Brain MR is a post-processing image analysis software that assists clinicians in viewing, analyzing, and evaluating MR brain examinations for the assessment of the volumetric properties of a variety of brain structures and white matter hyperintensities.

2.1.1.3 Indications for Use

AI-Rad Companion Brain MR is a post-processing image analysis software that assists clinicians in viewing, analyzing, and evaluating MR brain images.

AI-Rad Companion Brain MR provides the following functionalities:

- Automatic segmentation and quantitative analysis of individual brain structures and white matter hyperintensities
- Quantitative comparison of each brain structure with normative data from a healthy population
- Presentation of results for reporting that includes all numerical values as well as visualization of these results

2.1.1.4 Contraindications for Use

There are no known contraindications for the use of this product.

2.1.1.5 Limitations of Use

Feature: Brain Morphometry

The following limitations apply for the use of the Brain Morphometry feature:

- Non-T1-weighted MPRAGE series
- T1-weighted MPRAGE that have been acquired after contrast agent application
- Significant asymmetry between left and right hemispheres
- Image data from third party vendors that is not 3D distortion corrected
- Enhanced MR data format

Scan parameters of the MPRAGE protocol that are outside the required range according to the ADNI standard are not strict limitations. One can also allow result calculation for such non-ADNI compliant image data, but with possible loss of image quality.

Feature: White Matter Hyperintensities

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The following limitations apply for the use of the White Matter Hyperintensities feature:

- Non-T1-weighted MPRAGE or Non-T2-weighted FLAIR series
- T1-weighted MPRAGE that have been acquired after contrast agent application
- Image series acquired from other than Siemens Healthineers scanners
- Enhanced MR data format

2.1.1.6 Intended Patient Population

For the Brain Morphometry feature, the intended patient target group consists of patients of age two years or above. In this age range, the brain segmentation algorithm works properly.

The normative range database provides the knowledge to compare volumetric values of brain structures with an age-dependent model of a reference population. The model of the reference population is based on certain datasets that were used to create the model. Due to this model, the age range between 16 and 98 years of age is well covered by the background database. However, for patients under 16 years of age and older than 98 years of age, the model needs to extrapolate from the base information. As the extrapolation is not as accurate and validated as the use within the valid age range, it will not be offered.

Therefore, the volumetric data and segmentation results can be calculated for all patient data from two years of age onwards. However, the comparison with the reference population is available only for the age range from 16 to 98. If the patient age is below 16 or higher than 98, comparison with normative data and the generation of the deviation map is disabled.

For the White Matter Hyperintensities feature, the intended patient target group is of age higher than 18 years, hence the results will not be offered for the patient age below or equal to 18 years.

2.1.1.7 Intended Users

AI-Rad Companion Brain MR is intended for the following user groups:

- Radiologist
- Clinical IT Administrator (including technologists)

User Role	Tasks Required	Competency Profile
Radiologist (clinical user)	• Reading of the results provided by AI-Rad Companion Brain MR • Archiving of results in PACS • Configuration of some application settings (e.g. clinical workflow with or without confirmation step)	• Radiologist • Professional education: Radiology or equivalent professional degree • Experience with PACS/RIS systems and basic IT knowledge • Confidence in the use of SW applications that are in scope of work • Ability to learn and adapt to a new device and willingness and motivation to learn to use a new device
Clinical IT Administrator (incl. technologists)	• IT related configuration setting (e.g. input and output target nodes, management of institution roles and user privileges).	• Professional education in IT • Experience with PACS/RIS systems and IT knowledge and/or medical IT network responsible assigned by the healthcare

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	• Support of the healthcare professional to process the data (configuring routing and processing rules)	organization

Reference for all 2.1.1 section: Market Requirements Specification [7]

2.1.2 Product Identification, Classification and GMDN Code

Material Number	Basic UDI-DI for product family	Product Name / Trade Name	GTIN/ UDI-DI	Classification acc. to Annex VIII EU MDR	Rule
11582982	0405686902277VW	AI-Rad Companion Brain MR	04068151011432	Class IIa	11a

MDR Codes:

Code	Term
MDA: 0315	Software
MDT: 2012	Devices which require installation, refurbishment
MDT: 2010	Devices manufactured using electronic components including communication devices

EMDN (CND) Code:

EMDN Code	EMDN Term
Z12040192	General Medicine Diagnosis and Monitoring Instruments – Medical Device Software

GMDN Definitions:

GMDN Code	GMDN Term & Definition
57812	Term: Radiology DICOM image processing application software Definition: A software application intended to facilitate management, distribution, and/or viewing of radiological digital imaging and communications in medicine (DICOM) image data; it is not intended for pathological image interpretation (i.e., not intended to detect and localize suspected abnormalities). It may be intended to facilitate image management (e.g., viewing, communication, printing, recording/disc burning) and/or image manipulation and quantification (e.g., non-interpretive graphical overlays).

Note: product is released for general availability with first SW full version VA50A. In case of sub-versions or any Service Pack / hotfix, only SW full version (UDI-PI) is modified, and UDI-DI remains the same.

Reference Document: Product Labeling Specification [78] and Regulatory Compliance Plan [13].

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2.1.3 Justification for product being a medical device

In order to properly classify a SaMD in accordance with the European Medical Device Regulation (MDR) [1] the guidance document MDCG 2019-11 [2] defines the criteria for the qualification and classification of software.

Guidance MDCG 2019-11 [2] establishes that in order to be qualified as medical device software, the product must first fulfil the definition of software according to this guidance and the definition of a medical device according to Article 2(1) of Regulation (EU) 2017/745 – MDR. [1].

Same guidance provides in Figure 1 a workflow as guidance for the decision steps for qualification of software as MDSW. These decision steps help arrive to the conclusion that AI-Rad Companion Brain MR qualifies as a medical device software:

- Step 1: Yes. AI-Rad Companion Brain MR is a software as per the intended use statement, it processes input data (i.e. DICOM / MR Brain images) and creates output data, therefore meets the definition of "software" according to Section 2 (Definitions and Abbreviations) of this guidance, as a set of instructions that processes input data and creates output data.
- Step 2: No. AI-Rad Companion Brain MR is a device in its own right, not specifically intended to be used together with a "particular" PACS or reading workstation to specifically enable them to be used as intended by the manufacturer, neither intended to specifically and directly "assist" the medical functionality of the medical device(s) in terms of its/their intended purpose(s); and it is therefore not deemed to be an accessory. AI-Rad Companion Brain MR is a standalone software which creates information on its own for particular medical purpose and performs its medical purpose without being part of a hardware medical device or any other Radiology Information Systems, nor drives or influences the use of another (hardware) medical device. Additionally, the AI-Rad Companion Brain MR software creates information on its own for particular medical purpose and does not drive or influence the use of other (hardware) medical device, nor its intended use corresponds to any of the device types listed in the MDR Annex XVI.
- Step 3: Yes. AI-Rad Companion Brain MR is a post-processing image analysis software that assists clinicians in viewing, analyzing, and evaluating MR brain examinations for the assessment of the volumetric properties of a variety of brain structures and white matter hyperintensities. AI-Rad Companion Brain MR is aimed to process images and provide quantifiable information to support diagnosis evaluations, and performs therefore an action beyond storage, archival, communication, simple search, lossless compression
- Step 4: Yes. AI-Rad Companion Brain MR outputs are considered to aggregate relevant clinical information that will not trigger an immediate or near-term clinical action (i.e. informative or adjunctive), but could be used for the clinical evaluation of individual patients as part of the medical examination.
- Step 5: Yes. As per intended use the AI-Rad Companion Brain MR qualifies as medical device software (MDSW) according to the definition of the guidance.

Given the definitions in Article 2(1) of MDR [1], the decision steps for qualification of software as MDSW (steps 1 to 5 in Figure 1 in the guidance) from MDCG 2019-11 [2] and the intended use of AI-Rad Companion Brain MR, it can be concluded that **it qualifies as a Medical Device Software according to EU MDR** [1].

Reference: Regulatory Compliance Plan (section 4.1) [13].

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2.1.4 Classification Rule and Rational for Decision

AI-Rad Companion Brain MR is a medical device software that achieves its intended purpose independent of any other device and therefore will be classified in its own right.

Software is defined as an active device and for the classification of active devices, Rules 9, 10, 11, 12, 13, 15 and 22 of Annex VIII of MDR [1] may apply.

Rule 22 applies for closed loop systems and Rule 15 for devices used for contraception. Rules 9, 10 and 12 are also not applicable considering the intended use of the product.

Rule 11, however, applies to Software intended to provide information which is used to take decisions with diagnosis purposes (Sub-rule 11a [1]). Exceptions of this sub rule are not applicable since the intended use of this software is limited to "*inform clinical management*" for a healthcare situation or conditions addressed as "*non-serious*". The information provided by AI-Rad Companion Brain MR is considered to only provide clinical information by aggregating relevant information that will not trigger an immediate or near-term clinical action (i.e. informative or adjunctive), in situations or conditions where an accurate diagnosis is important but not critical for interventions to mitigate long term irreversible consequences on an individual patient's health condition. Therefore, decisions taken with information provided by AI-Rad Companion Brain MR are not expected to have an impact that may cause death or an irreversible deterioration of a person's state of health, nor a serious deterioration of a person's state of health or a surgical intervention.

By application of the Rule 11 (sub-rule 11a) of Annex VIII of MDR [1] and considering the MDCG 2019-11 [2] (see also Figure 1) as guidance for classification, the overall regulatory classification for **AI-Rad Companion Brain MR is class IIa.**

		Significance of Information provided by the MDSW to a healthcare situation related to diagnosis/therapy		
State of Healthcare situation or patient condition		**High** Treat or diagnose *~ IMDRF 5.1.1*	**Medium** Drives clinical management *~ IMDRF 5.1.2*	**Low** Informs clinical management *(everything else)*
	Critical situation or patient condition *~ IMDRF 5.2.1*	**Class III** *Category IV.i*	**Class IIb** *Category III.i*	**Class IIa** *Category II.i*
	Serious situation or patient condition *~ IMDRF 5.2.2*	**Class IIb** *Category III.ii*	**Class IIa** *Category II.ii*	**Class IIa** *Category I.ii*
	Non-serious situation or patient condition *(everything else)*	**Class IIa** *Category II.iii*	**Class IIa** *Category I.iii*	**Class IIa** *Category I.i*

Figure 1: Classification Table as per MDCG Guidance 2019-11

Reference document: Regulatory Compliance Plan (section 4.1) [13].

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2.1.5 Marketing History of the Device

Following product versions are the predecessor devices of the device under evaluation:

Product Name	UDI-DI (GTIN)	UDI-PI (SW full version)	Installed Base (Customer sites)
AI-Rad Companion Brain MR	MDD device, UDI-DI not applicable (GTIN: 04056869257945)	VA20A*	0
		VA20A_HF01*	0
		VA20A_HF02	0
		VA21A	0
	04056869274621	VA30A	0
		VA30B	0
		VA30C	6
	04056869298207	VA40A	n.a.**
		VA40B	0
		VA40C	45 (from which 25 under trial license)

Table 1: Marketing History

** Versions were available only within collaboration for Customer Use Test.*

*** Version not deployed*

Note: due to the cloud-based delivery of the product, during SW update older versions in each datacenter are completely replaced with the latest one and no installed base remains with previous versions. This table shows the complete installed base information on 5st July 2023.

2.2 Principle of Operation of the Device and Its Mode of Action

2.2.1 General description of AI-Rad Companion product family

AI-Rad Companion product family provides automated post-processing of medical images by integrating Artificial Intelligence (AI) algorithms. The results generated by AI-Rad Companion can be sent to predefined systems that can receive DICOM objects.

The different AI-Rad Companion clinical extensions, part of the AI-Rad Companion product family, are considered as separate medical devices on their own, with own intended use, and are responsible for the post-processing of the specific DICOM images and generating the intended clinical output. The teamplay digital health platform and the AI-Rad Companion Engine are considered as non-medical software applications which serve as the SW communication platform infrastructure for AI-Rad Companion clinical extensions to run on.

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2.2.2 Principle of Operation

AI-Rad Companion Brain MR is a post-processing image analysis software that assists clinicians in viewing, analyzing, and evaluating MR brain examinations for the assessment of the volumetric properties of a variety of brain structures and white matter hyperintensities.

AI-Rad Companion product family provides automated post-processing of medical images by integrating Artificial Intelligence (AI) algorithms. The results generated by AI-Rad Companion can be sent to predefined systems that can receive DICOM objects.

The different AI-Rad Companion clinical extensions, part of the AI-Rad Companion, are considered separate medical devices on their own, with own intended use, and are responsible for the post-processing of the specific DICOM images and generating the intended clinical output.

2.2.3 Typical Clinical Workflow

A MRI head examination acquired as standard of care typically includes contrasts such as T1, T2, T2* and DWI contrast. The variety of contrasts is required to rule out other pathologies by the radiologist, such as intracerebral lesions. The T1 contrast with its good white matter / gray matter image contrast is used for the assessment of the volumetric properties of the various brain structures of interest. To detect white matter hyperintensities, the T2 contrast is also taken into account. During the scan a first quality assessment of the data is carried out to identify acquisition related problems (e.g. artefacts/motion) and to order repeat acquisitions if required.

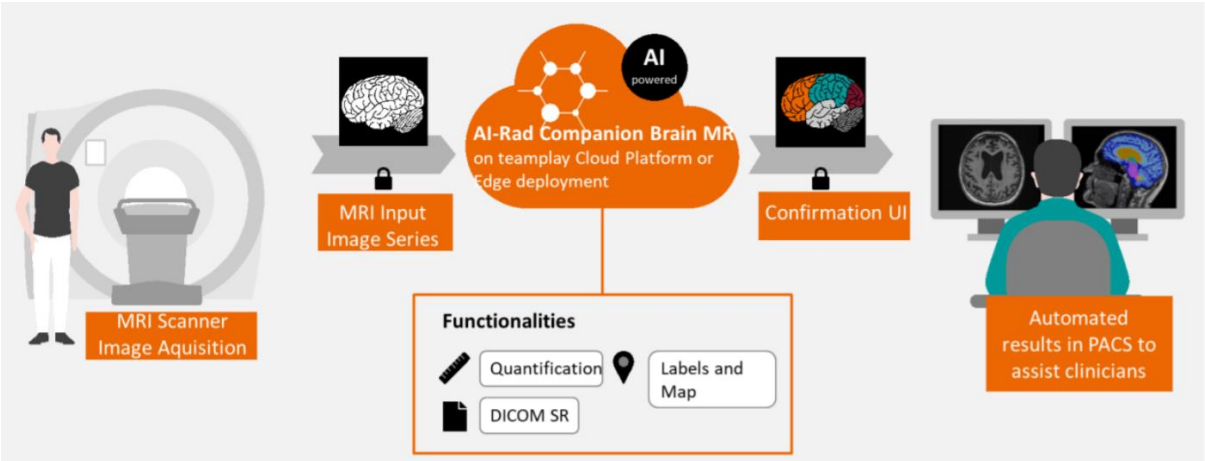

Figure 1: General workflow of the AI-Rad Companion Brain MR

In a typical standard of care, the images are archived in the PACS after image acquisition and read by the radiologist. The radiologist typically visually assesses the brain structures and extent of WMH and may use some measurement tools. The manual segmentation and quantification of brain structures and WMH is very tedious and time consuming, hence not part of the typical clinical routine. The segmentation and quantification tasks can be automated, and results be made readily available in PACS as an aid to the radiologists.

2.2.4 Device Description

The following use scenarios are supported by AI-Rad Companion Brain MR:

– Automatic segmentation and volume quantification of supported brain structures

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- Automatic comparison of volumetric values against a database of healthy reference subjects
- Automatic segmentation and quantification of White Matter Hyperintensities (WMH)
- Automatic results exporting to PACS
- Results exported in DICOM SR format

As part of the general workflow of this product, after the acquisition has ended, the relevant data is archived in the PACS and also sent to AI-Rad Companion for further processing. AI-Rad Companion Brain MR receives the study and automatically processes the data to generate the results (Figure 1). The user has the option to individually select the features of brain morphometry and/or white matter hyperintensities through the configuration UI.

The following two workflow options are supported:

- Semi-automatic workflow: All results need to be reviewed by the user in a dedicated user interface prior to being sent to PACS, which is marked as confirmation UI in Figure 1. The user has the possibility to decline results, and only accept those results that shall be sent to PACS. This is the default workflow.
- Automatic workflow: The user can alternatively configure the application to allow sending results automatically to PACS. This is an optional workflow.

The following results generated by AI-Rad Companion Brain MR aid the radiologists in reading the case in the PACS.

A. <u>Results generated by the Brain Morphometry feature</u>:

- Volumetry report: Table of all relevant volumetric tissue class and brain structure values together with the segmentation quality metrics. The report includes volume estimates for the different brain regions. Both absolute and relative (percentage of total intracranial volume) volumes are reported. Additionally, for each brain region, a normative range is displayed and corresponds to the 10th and 90th percentiles of an age- and gender-matched healthy population.
- List of supported brain structures: Total Intracranial Volume, Whole Brain, Gray Matter (GM), Cortical GM, White Matter (WM), *Abnormal WM, Cerebo Spinal Fluid (CSF), Thalamus left/right, Putamen left/right, *Caudate left/right, Pallidum left/right, Hippocampus left/right, Lateral ventricle left/right, 3rd ventricle, 4th ventricle, GM frontal lobe left/right, GM parietal lobe left/right, GM occipital lobe left/right, GM temporal lobe left/right, GM cingulate gyrus left/right, Insula left/right, WM frontal lobe left/right, WM parietal lobe left/right, WM occipital lobe left/right, WM temporal lobe left/right, Cerebellum (WM + GM), Mesencephalon, Pons, Medulla Oblongata, Corpus Callosum (midsagittal area)

  **Note: Caudate left/right and Abnormal WM are not available for the USA. The following structures are covered*:

- Label map: Shows the different brain regions, each region in a different color
- Label map fused with the MPRAGE image (color-coded and partially blended)
- Deviation map: Shows the deviation for each brain structure from an average age and gender-matched normative volume. A color-coded representation of the deviation map is used which allows the assessment of the deviation from the mean normative values for each tissue class at one glance.
- Deviation map fused with the MPRAGE image (color-coded and partially blended)

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- DICOM Structured Reporting output format

B. Results generated by the White Matter Hyperintensities (WMH) feature:

- WMH map visualizing the white matter hyperintensities
- WMH map visualizing the white matter hyperintensities fused with the registered T2 FLAIR image (color-coded and partially blended)
- Report: Table summarizing all numeric results, indicating the total count and volume of the WMH, as well as the WMH volume for each of the four subregions periventricular, juxtacortical, infratentorial, and deep white matter
- DICOM Structured Reporting output format

Archiving is not covered by AI-Rad Companion and must be performed in the PACS.

Reference and further details can be found in Market Requirements Specification [7].

2.2.5 Operation Environment and Installation

2.2.5.1 Operation Environment

AI-Rad Companion Brain MR is designed to be used in a healthcare environment at a radiological workplace or on a workstation used for reporting or interpretation of MR brain images, with restricted access for authorized personal only.

Frequency of use can be several times per day and depends on the hospital's patient load.

AI-Rad Companion Brain MR is not validated for use with touch screen or mobile devices.

Reference: Market Requirements Specification [7].

2.2.5.2 Installation

AI-Rad Companion Brain MR supports cloud-based and on-edge (on-premises, connected to cloud) deployment models and is operated by Siemens Healthineers. This enables the institution to select the option that suits their requirements.

AI-Rad Companion is deployed through teamplay digital health platform. If the customer already has the Siemens Healthineers cloud infrastructure in place, AI-Rad Companion do not require any installation, migration, or implementation. Therefore, there are no installation steps nor instructions to be provided to customers in scope of AI-Rad Companion devices.

In the case of Cloud deployment, the computer hosting the teamplay Receiver requires access to the hospital intranet and the internet with at least 100Mbit/s bandwidth to the internet

In the case of Edge deployment, the patient data, queueing, database, storage, and execution environment for the processing algorithms will be on-premises but still login, key vault, institution management, auditing, logging will be in the cloud. In edge deployment, data processing is performed within the institution and therefore, minimum hardware requirements are also provided in the device's Instructions for Use [79]. It is the responsibility of the customer to provide and maintain such (off-the-shelf) hardware as well as securing the on-premises installation (e.g., physical access control).

Software update and upgrade packages are made available in the cloud and are installed automatically. Thus, AI-Rad Companion updates and upgrades do not have to be installed by the user or customer's IT Administrator. In this sense, only internal guidelines and instructions

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for deployment team are issued within the document for internal use only, Installation and Deployment Instructions [86], describing the set of activities to be performed by Siemens Healthineers' employees directly, during new or update of deployments part of software rollouts.

2.2.6 AI Components of the Product

The AI-Rad Companion Brain MR runs two distinct and independent algorithms for Brain Morphometry analysis and WMH segmentation, respectively.

In overall, comprises of 2 main algorithmic features:

- Brain morphometry
- White Matter Hyperintensities (WMH)

The *Brain Morphometry feature* uses an algorithm that performs an atlas-free tissue classification (GM, WM, CSF) followed by an atlas-based brain structure segmentation (e.g. thalamus, putamen, hippocampus, etc.) to estimate volumes. The atlas-free tissue classification is based on a classic machine learning method, expectation maximization, that computes an accurate estimation of the means of the main tissues based on prior information in the brains of several healthy subjects. The atlas-based segmentation algorithm is based on expert manual segmentation of a template image. This algorithm pipeline has not been changed since the VA2x mainline and therefore the scientific evaluation VA21 [70] remains valid for Brain Morphometry.

This type of algorithm does not require training data (i.e. it does not learn from data to make predictions) and data used to build the atlas algorithm cannot be considered as training data.

For Brain Morphometry feature, details on the applicable testing methods and testing data used as part of the algorithm development can be found in the scientific evaluation document [70].

The new *WMH feature* introduced within this mainline version uses an algorithm that performs an initial segmentation based on a k-nearest neighbors classifier refined by an additional estimation of the partial volume in surrounding areas. A deep learning-based classifier is then used to reduce the false positives in the initial WMH segmentation. These learning algorithms were trained using expert manual segmentations of WMH from FLAIR image series.

All algorithms are locked and do not change during the product use. The product does not provide any automatic diagnosis or treatment decisions, but only provides results to the radiologists as an aid in assessing Brain MR images.

All other relevant details for the algorithm development, as well as the training and testing data handling, for WMH feature, can be found in the corresponding Quality Plan for AI/ML [17], Data Management Plan [18] and specific Scientific Evaluation [71].

2.2.6.1 Architectural Design

Refer to Design Specification documents (for Brain MR and specific for WMH algorithm workflow [22][23]) for detailed design description.

2.2.6.2 Algorithm Risk Assessment

Risk associated with use of AI algorithm(s) and risk benefit consideration is evaluated in the risk management file and clinical evaluation report [25][26][69].

Main algorithm (segmentation) accuracy and/or performance parameters are also traced within

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the Product Risk Analysis [25] deriving in traceable Software Requirements [8].

2.2.6.3 Algorithm integration, Verification and Validation

For the testing of algorithm within the application, several test levels are also perfumed.

For the algorithm integration & verification (Morphometry & WMH), refer to Scientific Evaluation Reports [70][71] and Integration Report [50].

For the algorithm validation (Morphometry & WHM), refer to System Test Report [51] and System Validation Report [52], where the test cases cover the algorithm validation implicitly.

2.2.7 Accessories and device combinations

AI-Rad Companion Brain MR is used in combination with AI-Rad Companion Engine, which is a non-medical device providing the communication platform and the means for storing and transferring data into other systems only. AI-Rad Companion Brain MR it is solely responsible for the medical functionality according to its intended purpose.

A detailed evaluation under MDR Art.22 was performed in the Regulatory Compliance Plan [13] section “determination of applicability of Article 22 of the EU MDR 2017/745”, concluding that use of AI-Rad Companion Brain MR on the AI-Rad Companion Engine, does not fulfil the definition for an “Accessory” nor for a “System” as per definition in MDR Art 2(11), and therefore a declaration under Art. 22 is not deemed as applicable.

GSPR 14.1 applies, and the compatibility and interoperability of AI-Rad Companion Brain MR and AI-Rad Companion Engine is verified during the Verification and Validation activities [13][36][53][55][59][67]. Information about the compatible versions of the AI-Rad Companion Engine are also described in Section 1 of the Instructions for Use [79].

2.2.8 Description or complete list of the various Configurations and variants of the device

There are no various configurations or variants of AI-Rad Companion Brain MR.

2.2.9 Description of novel features

AI-Rad Companion Brain MR does not include any novel features.

2.2.10 General Description of the Key Functional Elements

Covered as part of the principle of operation and mode of action, section 2.2 .

2.2.11 Materials incorporated into Key Functional Elements

Not applicable. Device is a medical device software.

2.2.12 Technical Specifications

Details on technical specifications can be found in the Instructions for Use [79].

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2.3 Previous and Similar Generations of the Device

2.3.1 Previous Generations of the Device

Please refer to marketing history, section 2.1.5

2.3.2 Identified Similar Devices available in the European Union

Identified similar devices:

- Icobrain (by Icometrix)
- NeuroQuant™ (by CorTechs Labs inc)
- QUANTIB®Brain (by Quantib B.V.)

See detailed comparison in PMCF plan [74].

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3 Information supplied by the Manufacturer

3.1 Product Label(s)

All applicable label requirements are covered with the About Box since this device will be delivered digitally, without any physical media. About box is translated for all the languages (including those required by the EU member states) supported by the graphical user interface.

See example below .

Figure 2 Example of About Box AI-Rad Companion Brain MR VA4

References: For details about label content refer to Product Label Specification [78]. For further information on UI supported languages, refer to section of Target Countries and Languages in MRS [7].

3.2 Packaging Labels

All packaging labeling requirements are covered with the About Box since this device will be delivered digitally, without any physical media.

Reference: see example in Figure 1 and Product Label Specification [78] for details.

3.3 Instructions for Use

AI-Rad Companion Brain MR is delivered with Instructions for Use in electronic form (within the application in form of online help), according to EU Commission Regulation on electronic instructions for use of medical devices (EU) Nr. 2021/2226 [3][95][96]. In general, information to be provided to user regarding use of electronic instructions for use and ways of access are covered within the about box of the device.

Instructions for Use are made available within the SW itself in form of Online Help and additionally as pdf. document through Siemens Healthineers Document Library, which access is restricted to SHS customers.

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Target countries are part of the Market Requirements Specification [7] and applicable language requirements are assessed within the RCP [13]. Considering that AI-Rad Companion is hosted by teamplay digital health platform, first set of potential target markets were selected based on teamplay availability, and priorities are handled depending on business opportunities. In general, the initial assessment is always considered for main target markets EU and USA and other MDSAP countries, and additional countries will be further added during product life cycle on request.

For Instructions for Use in English and available translations in languages accepted in the EU Member States and other countries where the device is envisaged to be sold, see User Documentation Plan (UDP) [6] and Release for Distribution (R4D) [82] documents.

Translations are verified by a Native Language Specialist (NLS) and documented via NLS approval records [81].

Additionally, a digital leaflet as an Addendum document in pdf. [84] is provided to the user with detailed instructions on how to get started and different ways to access the Instructions for Use.

3.4 Other documents supplied to user

The Release Information document [85] provides an overview for the customer on the content of the version released, including information of fixed problems, new features or known limitations.

A Datasheet document [83] is issued, that serves as a presales deliverable that can be handed over to customers or attached to tenders after product release.

Instructions for Use of the AI-Rad Companion Engine as the SW platform are also available for users [80] in same electronic options (online help or document library) and same language translations.

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4 Design and Manufacturing

4.1 Design Stages applied to the device

AI-Rad Companion Brain MR is developed in accordance with the Siemens Healthcare GmbH, Digital Health Quality Management System (DH QMS), Design Control Process DH QR 04 [98].

The PLM process development model according to DH QR 04 is divided into several phases. At the synchronization point/Main Review (R1, R2, R3, R-Sys, R-CUT, R4, R5), the defined quality criteria are assessed.

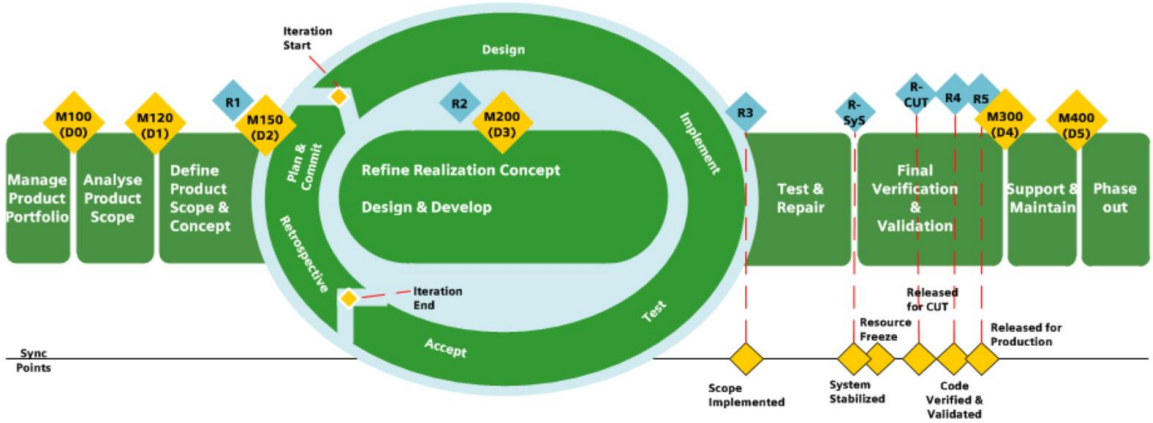

Figure 3: Process model with Phases, Milestones and Main Reviews

Az the Milestones/Decision points (M100, M120, M150, M200, M300, M400), the Management Board considers the results of the previous phase and the strategic project and business situation to decide about the start of the next phase.

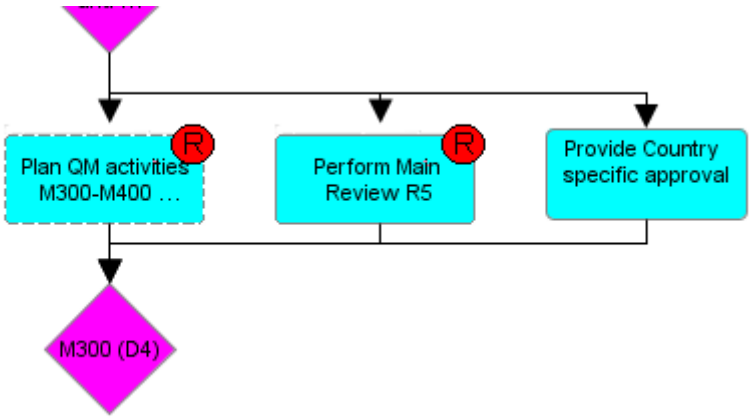

Figure 4 Quality Management Discipline (DH PLM D 116) - Main Review R5

The Main Review R5 represents the release for Production and indicates that the device is released.

As part of the Main review R5, all GSPR requirements as required by MDR are checked and conformity is confirmed by the RAM, PM and QM by filling and signing the GSPR Checklist. After confirming compliance to GSPR [92], the Declaration of Conformity [90] is signed by

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the Head of Business Line and the Quality Management Representative. Supply chain is established to support the final delivery.

The release of the product for general availability is communicated and approved by D&A Management at M300 milestone.

By following the Design Control Process DH QR 04 [98], AI-Rad Companion Brain MR development fulfills the requirements of the IEC 62304 [64], IEC 62366-1 [66] and 82304-1 [65].

4.1.1 Cybersecurity

A Cybersecurity Concept was defined for AI-Rad Companion family [30].

Cyber Security requirements are addressed following the DH QMS Secure Development Lifecycle process for Product and Solution Security (PSS) [99][108][109].

The PSS Classification for each medical device is documented in the PSS classification document [27] and respective MRS [7]. This device has been classified with a *PSS high*, refer to PSS Classification Extensions document [27] for details and where applicable, refer to comments labelled as “MR Brain”.

Based on the PSS Classification, security activities are planned according to the Instruction 15_I_25 [109].

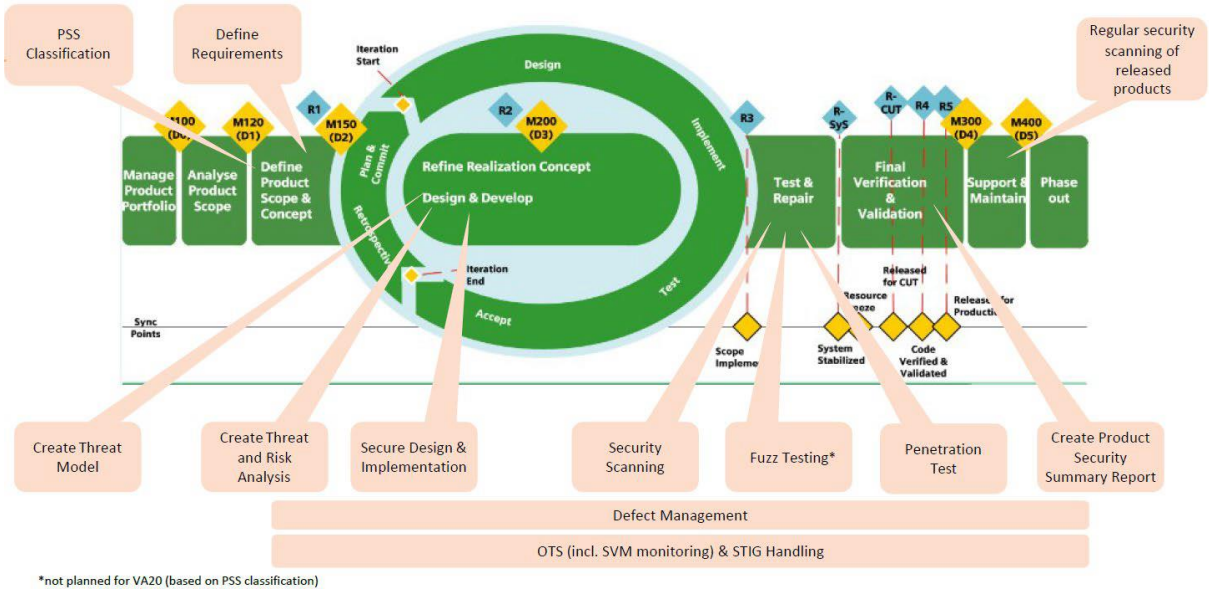

Figure 5 –PSS activities

A high-level system overview of the AI-Rad Companion family (including medical and non-medical devices and interfaces to other products) is available in the System Threat and Risk Analysis Report [28]. Please note that the System Threat and Risk Analysis Report is applicable for the whole AI-Rad Companion product family which consists of several software products medical devices under their own right.

Cybersecurity Risk Assessment:

The PSS Classification for each medical device (Extensions) is documented in AI-Rad Companion PSS Classification-Extensions [27] document and the respective Market Requirement

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Specifications [7]. Based on the PSS Classification-Extensions, security activities are planned according to the SHS QT QR 25 Appendix A.2 [109].

Cybersecurity risks are being analyzed and cybersecurity risk control measures are being defined in the Threat and Risk Analysis Report. Cybersecurity risk control measures applicable for AI-Rad Companion Brain MR are traceable through the SWR keys mentioned in the "System Threat and Risk Analysis Report" [28].

A final comprehensive report on the cybersecurity aspects of the product and residual cybersecurity risks is available in the Product Security Summary Report (PSSR) [29].

Accompanying Documentation:

Cybersecurity information is provided to the user in the Instructions for Use of AI-Rad Companion Brain MR [79], Instructions for Use of AI-Rad Companion Engine [80] and the Security Whitepaper and MDS^2 [30].

Software Bill of Material and SOUP/OTS components:

The OTS and supplied software catalog includes a list of all commercial, open-source, and off-the-shelf software components including SOUP. Please refer to [24].

Cybersecurity Testing

Several type of security testing are implemented within AI-Rad Companion devices, like penetration testing (incl. fuzz testing) [34], vulnerability scanning (SSCAN) [35], as well as participating in a bug bounty program established within D&A. Results are recorded and summary of results is part of the Product Security Summary Report (PSSR) [29].

Post-Market Surveillance:

The PMS Plan [73] covers cybersecurity aspects for the Post-Market Surveillance. Records for post-market cybersecurity activities are available in applicable PMS report(s) [75] (as specified in PMS Plan) and/or summarized in the PSUR [77].

For additional details on cybersecurity aspects of AI-Rad Companion Brain MR, please refer to the project documents in section 13.6 including available whitepapers.

Since 2022, AI-Rad Companion is also certified as fulfilling the requirements according to ISO/IEC 27001:2013 + Cor1:2014 + Cor2:2015, regarding provision of secure design, development and operation.

4.1.2 Data Privacy

A Data Privacy Concept was defined for AI-Rad Companion family, and this is described in the Data Privacy Concept document [31] and Data Privacy and Security White Paper for AI-Rad Companion [33]. It provides an overview of how the architecture, configuration, and administration of AI-Rad Companion ensures "Privacy by Design".

AI-Rad Companion family of products is registered in the Healthineers Data Privacy Register (HDP) and the updated valid registration can be found under "AI-RAD Companion" or "584k-12539-5.00" registration number.

Please refer for details to: Data Privacy Concept [31] and Data Privacy and Security White Paper [33]

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4.2 Packaging, Storage and Distribution

AI-Rad Companion Brain MR is a cloud-based software only application. It means, distribution is made by Digital Delivery Service and not by delivering SW in any physical media (e.g. CD, DVD), therefore, packaging and storage activities do not apply in this case.

Digital Delivery considers activities regarding offer and order processing to ensure that customer can access the services, covering enabling according to country approvals and tracking of entitled services. Distribution activities are being planned and conducted according to established procedures.

The operation of AI-Rad Companion is developed according to the AI-Rad Companion Operations Development Plan [15] and finally established in the Operations Concept [16].

Refer to the following documents for the process framework:

- DH QR 29 Operations [105]
- DH PLM D 133 Digital Delivery [106]

4.3 Design and Manufacturing Specifications

For Design Specifications see Design documents referenced in Sections 13.2 13.3 13.4 13.5 13.6 .

Considering this is a software only product to be delivered digitally, there is not a manufacturing process as such applicable. Final software build released is transferred to a separate environment, a design transfer test record [72] is generated and software is made available for the next logistic and operation process.

4.4 Identification of all Sites incl. suppliers and sub-contractors

4.4.1 Design and Manufacturing Sites

4.4.1.1 Manufacturer

Siemens Healthineers AG
Siemensstr. 3
91301 Forchheim
Germany

4.4.1.2 Authorized Representative

Not applicable, manufacturer is located in the European Union.

4.4.2 Facility

Siemens Healthineers AG
Digital & Automation (D&A)
Henkestr. 127
91052 Erlangen
Germany

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4.4.3 Relevant suppliers and sub-contractors

Following critical suppliers and sub-contractors are identified:

Key functional element	Relevant suppliers and sub-contractors	Activities (Design and/or Manufacturing)	Testing at supplier/sub-contractor	Control measures (verification and/or validation, incoming Inspection)	Manufacturing test described in manufacturing instruction
Various design and development activities for MDSW	Siemens Healthcare Private Limited. Bangalore, India	Software development services (Development activities)	V&V and clinical evaluation performed acc. DH QMS.	- Project individual agreement. - Supplier's ISO 13485 certificate* - Milestones acceptance (details in 4.1) /R5 milestone [94]	Not applicable.

**Supplier's certificates are handled / archived by Supplier Management and can be provided on request.*

A detailed list of suppliers and supplier management is provided in the Project Management Plan [10].

Details on supplier management are provided in the Project Management Plan [10]. New Algorithm (WMH) is developed internally, therefore no changes in the list of critical suppliers applicable in scope of this device version.

Note:

SHS defines "*relevant suppliers and sub-contractors*" in accordance with the definitions of the term "*Critical Supplier*" in ZLG 3.9 B17 and NBOG BPG 2010-1 documents and TÜV Document "*All you need to know about unannounced audits – Frequently Asked Questions* (V-M/MHS/8.1/SG)"

Thus "*relevant supplier and sub-contractors*" are understood as follows:

A "*Relevant supplier or sub-contractor*" is a supplier or / and sub-contractor:

- Providing Original Equipment Manufactured products (OEM);
- Involved in the design and development of medical devices or software (on integration test-level without verification and / or validation by Siemens Healthineers unit);
- Providing processes which require a validation, such as sterilization, sterile packaging, virus inactivation; OR
- Providing critical materials, in the meaning of components with high integrity characteristics (HIC) (Defined in EN 60601-1: 2005).

4.4.4 Other Economic Operators

4.4.4.1 Importer

N/A, manufacturer is located in the European Union.

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4.4.5 Person Responsible for Regulatory Compliance

Head of Quality for Business Line Digital & Automation (SHS DI D&A) is the Person Responsible for Regulatory Compliance (PRRC). Further details can be provided upon request.

4.5 General Safety and Performance Requirements (GSPR)

To demonstrate conformity with the General Safety and Performance Requirements, a separate checklist document is created, see GSPR Checklist [92].

The evidence regarding the conformity with GSPRs is based on harmonized standards, Common Specifications (CS) or other applicable (inter)national Standards. Furthermore, the design verification and validation documents are used.

Considering that at the time of release of the mainline version of the device, there is no publication of a full list of new harmonised standards applicable for EU MDR by the European Commission, and only applicable standards EN ISO 13485 and EN ISO 15223-1 and EN ISO 14971 were already harmonized within Decision (EU) 2022/6 (amending Decision (EU) 2021/1182), other harmonized standards applicable for MDD were also applied.

The revision and location (e.g. eDoc ID) of all documents which are referenced as an evidence in the GSPR Checklist can be found in the product and version specific Document Plan [4].

4.6 Benefit-Risk Analysis and Risk Management

Risk management is performed according to the relevant procedure, DH QR 16 [103]

Benefit-Risk analysis and solutions/results of Risk Management are covered as part of the documents listed in section 13.5 .

Analysis of risks related to use error are covered in the same Risk Management File (see Section 13.5 .). In general, compliance to usability standard is detailed in the applicable IEC 62366-1 Compliance Report [66] and corresponding records [60][61][62][63].

4.7 Product verification and validation

The evidence regarding the product verification and validation includes all relevant testing documents available under Design Verifications and Validations in section 13.7 .

For documentation with regards to the applicable GSPR see GSPR checklist [92].

4.7.1 Pre-clinical and clinical data

See below details on the types of tests performed or non-applicability of certain test types for AI-Rad Companion Brain MR is provided below (in line with MDR, Annex II, Clause 6.1).

Test type	Evidence
Animal tests	Not applicable, device is a MDSW
Biocompatibility	Not applicable, device is a MDSW

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Physical, chemical and microbiological characterization	Not applicable, device is a MDSW
Electrical safety and electromagnetic compatibility;	Not applicable, device is a MDSW
Design Verification	Relevant test documents: • Test concept (test planning document) [12]. • Test specifications (unit, integration system), see from [36] to [41] • Test list (integration, NFR, system, sanity), see from [42] to [46]. • Test Records (unit, integration, system, Interoperability, Performance, Sanity, DICOM): see from [47] to [55]. • Test Summary Record [56] & Trace document [57]
Simulated Use/Design Validation	• System Validation Test Specification [41] • System Validation Test List [44] • System Validation Test Record [52]
Usability Testing	• Customer Engagement Plan [60][61] • Please refer to [61][63] for formative and summative records for usability
Test and Compliance reports for specific standards	• IEC 62304 compliance report [64] • IEC 62366-1 compliance report [66] • IEC 82304-1 compliance report [65] • DICOM Conformance statement [67]
Directive 2004/10/EC relating to the application of the principles of good laboratory practice and the verification of their applications for tests on chemical substances	Not applicable, device is a MDSW
Stability Tests	Test concept [12] section 6.5 covers "*load tests*" which serve the purpose of testing stability, as well as 6.11.1 defined that non-functional requirements (NFRs) are also to be tested to validate the system meets the non-functional requirements including stability of the entire system. Load tests are covered as part of overall performance test [54].
Shelf life of device	Device is a software product only, hence "shelf life" requirements, as a date until when a product shall be

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	used, do not apply as such. For the lifetime of the device, see MRS [7] §2.11*: the lifetime of AI-Rad Companion Brain MR is estimated to be one year or less until a new version of the software is released. In each single market, customers will be automatically upgraded to the latest version when introducing a new software into the market due to the cloud nature of the product.* *The lifetime of the AI-Rad Companion product line after the last planned software release (feature release) is defined as 5 years, so SHS continues to support released product for this lifetime.*
Performance and Safety	Performance Tests: refer to Test concept [12] for details on performance test planning, and [54] for performance test results. Safety tests: safety requirements are part of SWRS and tests covered as part of overall system testing ([40][43][51]). Coverage of all risk aspects considered in the product design is available in the Risk Management Report [26]..
Clinical Evaluation (including evaluation of published literature)	• Clinical evaluation plan [68] • Clinical evaluation report [69]
Post Market Clinical Follow Up	• Post Market Clinical Follow-up Plan [74] • Post Market Clinical Follow-up Report [76]

4.7.2 Additional information required in specific cases

This technical documentation does not include information required in the following specific cases (not applicable to the device in scope):

a) Incorporated substances considered to be a medicinal product (incl. derived from human blood or human plasma)

b) Tissues/Cells of human or animal origin, or their derivatives (incl. materials of biological origin)

c) Nanoparticle technology

d) (Combinations of) Substances intended to be introduced into the human body and that are absorbed by or locally dispersed in the human body,

e) CMR or endocrine disrupting substances

f) Sterile devices or devices with defined microbiological condition

g) Infection risk and reusable device(s)

h) Devices with a measuring function

i) Devices with connection to other device(s) in order to operate as intended

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5 Post Market Surveillance

For AI-Rad Companion Brain MR, a post-market surveillance system has been established to actively and systematically gather, record and analyze relevant data on the quality, performance and safety of the device.

It includes:

- The device's Post Market Surveillance Plan which defines the systematic process for the collection and utilization of available post market data and information and references the appropriate processes and procedures for the notification, evaluation and reporting of Serious Incidents and Field Safety Corrective Actions (FSCA) as well as for the detection and reporting of trends of non-serious incidents thereby ensuring that the manufacturers obligations in regards to vigilance as laid down in MDR Articles 87, 88 and 89 are fulfilled and
- Post Market Surveillance records, will be generated separately in the established frequency, as described in the Post Market Surveillance Plan.

Technical documentation on Post Market Surveillance is updated throughout the entire life cycle of the device thereby ensuring that necessary conclusions are drawn, and any required preventive and corrective actions are determined, implemented and monitored appropriately. The PMS activities (plan and records) cover cybersecurity aspects for the Post-Market Surveillance as well.

As the device described in this document is a Class IIa device, periodic safety update reports (PSUR) will be/is created and updated at least every two years and archived as part of the technical documentation.

Please refer to: Post Market Surveillance Procedure (DH QR 31) [107] and specific PMS Documents (Section 13.11).

5.1 Post Market Clinical Follow Up

Post Market Clinical Follow Up (PMCF) activities for AI-Rad Companion Brain MR shall be carried out as established in the PMCF Plan [73]. Results of the plan shall be captured in a corresponding report, at a frequency as specified in the PMCF plan. If PMCF activities are not conducted for a specific release, justification for the same shall be captured.

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6 Other applicable EU legislations

There are no other applicable EU legislations for this software only device.

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7 QMS

- EN ISO 13485:2016 Certificate [88]
- MDSAP ISO 13485:2016 Certificate [89]
- EU Quality Management System Certificate (MDR) [87]

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8 Conformity Assessment Procedure

Applicable Conformity Assessment Procedure according to EU MDR:

ANNEX IX, Conformity assessment based on a quality management system and on assessment of technical documentation by a notified body.

Notified Body:

TÜV Süd Product Service GmbH
Ridlerstr. 65
DE-80339 München
Germany

Notified Body identification number: 0123

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9 EU Declaration of Conformity

The DoC for AI-Rad Companion Brain MR is issued covering the following EU legislation:

- EU MDR: (EU)2017/745 [1]

For the EU Declaration of Conformity document, see [90][92].

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10 Registration Data

Device registration modules in EUDAMED are not yet active at the time of release of this device, therefore device is not yet registered within EUDAMED. Until EUDAMED is considered fully operational, the existing requirements or local registration in Germany and notification in other EU countries shall continue to be followed.

Available product specific registration data for future registration in EUDAMED is prepared in the master data in SAP.

Manufacturer Single Registration Number (SRN): DE-MF-000038456.

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11 Design Changes

Design changes are controlled, evaluated and implemented in line with QR 15 [102]. Changes introduced to the device in scope of this document, including hotfixes, shall be summarized in a Scope of Update documents which will be added as part of the technical documentation (see section 13.15).

In general, hotfix versions (also called service packs) are only intended to restore the device to original specifications.

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12 GSPR Checklist

The GSPR checklist for AI-Rad Companion Brain MR is delivered as single record [92].

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13 List of all Documents Referenced

This list includes the most essential documents used to prove conformance to the GSPRs [92] and documents explicitly required by Annex II and III. These documents are typically submitted to Notified Bodies and National Competent Authorities in case the Technical Documentation is requested.

All documents of the Technical Documentation corresponding valid revisions are listed in the product specific Document Plan [4].

Please note:

For *AI-Rad Companion Brain MR* Technical Documentation, some valid documents are identified with the device specific name (further listed below as for *Brain MR*) and some documents may be identified as *AI-Rad Companion* only (further listed below as *AIRC*), for being valid for several devices part of the family of applications.

In case of documents identified as *AI-Rad Companion Cook* (listed below as *Cook*), this name identifies a "project planning" and such documents are valid for all independent applications developed under same project plan (refer to PMP for details [10]).

13.1 Documents Plans

[4]	DP	Document Plan AI-Rad Companion Brain MR VA40x	(latest released under VA40C_AR_LMT project)	
[5]	ODP	Operations Document Management Plan	AIRC	see [4]
[6]	UDP	User Document Plan	Brain MR	see [4]

13.2 Design Input Documents

[7]	MRS	Market Requirement Specification (MRS)	Brain MR	see [4]
[8]	SWRS	Software Requirement Specification (SWRS)	Brain MR	see [4]
[9]	UISG	UI Styleguide (SHUI)		see [4]

13.3 Design Planning Documents

[10]	PMP	Project Management Plan	Cook	see [4]
[11]	QMP	Quality Management Plan	Cook	see [4]
[12]	CIMP	Configuration & Integration Management Plan	Cook	see [4]
[13]	TC	Test Concept	Cook	see [4]
[14]	RCP	Regulatory Compliance Plan	AIRC Brain MR / see [4]	
[15]	OPS	Operations Development Plan	AIRC	see [5]
[16]	OPS	Operations Concept	AIRC	see [5]

<u>Algorithm development specifics</u>:

[17]	Q-Plan for AI-ML	Brain MR	see [4]
[18]	Data Management Plan	Brain MR	see [4]
[19]	Annotation Requirement Specification	Brain MR	see [4]

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13.4 Design Output Documents

[20]	SAS	System Architectural Specifications	AIRC	see [4]
[21]	SWAS	Software Architecture Specifications	Brain MR	see [4]
[22]	CDS	Design Specification	Brain MR	see [4]
[23]	CDS	Design Specification – Algo	Brain MR	see [4]
[24]	OTS	OTS catalog	AIRC	see [4]

13.5 Risk Management

[25]	RA	Product Risk Analysis	Brain MR	see [4]
[26]	RMR	Product Risk Management Report	Brain MR	see [4]

13.6 Cybersecurity and Data Privacy

[27]	PSS	PSS Classification-Extensions	AIRC	see [4]
[28]	TRA	System Threat and Risk Analysis (confidential)	AIRC	see [4]
[29]	PSSR	Product Security Summary Report (confidential)	Brain MR	see [4]
[30]	CONCEPT	Cybersecurity Concept	AIRC	see [4]
[31]	CONCEPT	Data Privacy Concept	AIRC	see [4]
[32]	CS	Security Whitepaper and MDS2	AIRC	see [4]
[33]	DPS	Data Privacy and Security Whitepaper	AIRC	see [4]
[34]	PEVAL	Penetration Testing Report (confidential)	AIRC	see [4]
[35]	SSCAN	Scanning Result Document (confidential)	AIRC	see [4]

13.7 Design Verification and Validation

Test Plans / Specifications:

[36]	TP	Test Plan DICOM Interoperability Test	Brain MR	see [4]
[37]	UTS	Unit Test Specification	Brain MR	see [4]
[38]	UTS	Unit Test Specification-Algo	Brain MR	see [4]
[39]	ITS	Integration Test Specification	Brain MR	see [4]
[40]	STS	System Test Specification	Brain MR	see [4]
[41]	SVAL	System Validation for Clinical Use	Brain MR	see [4]

Test Lists:

[42]	TL	Test List – IT (incl. for regression tests)	Brain MR	see [4]
[43]	TL	Test List – ST (formal and regression tests)	Brain MR	see [4]
[44]	TL	Test List - System Validation	Brain MR	see [4]
[45]	TL	Test List - NFR (also called Performance)	Brain MR	see [4]
[46]	TL	Test List - Sanity Test	Brain MR	see [4]

Test Records:

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[47]	UTR	Test Record - Unit Test	Brain MR	see [4]
[48]	UTR	Unit Test Records (Algo-CPP / Algo-Python)	Brain MR	see [4]
[49]	CRSR	Code Review Summary Record		
[50]	ITR	Test Record – IT (and regression)	Brain MR	see [4]
[51]	STR	Test Record - ST	Brain MR	see [4]
[52]	TR	Test Record - System Validation	Brain MR	see [4]
[53]	TR	Test Record - Interoperability Test (DICOM)	Brain MR	see [4]
[54]	TR	Test Record - NFR regression (also called Performance)	Brain MR	see [4]
[55]	TR	Test Report - Sanity Test	Brain MR	see [4]
[56]	TS	Test Summary	Brain MR	see [4]

Other test documents:

[57]	TRACE	Traceability Matrix	Brain MR	see [4]
[58]	UA	Unresolved Anomalies	Brain MR	see [4]
[59]	CM	Compatibility Matrix	AIRC	see [4]

Usability records:

[60]	CUEP	Customer Engagement Plan (VA40)	Brain MR	see [4]
[61]	CUEP	Customer Engagement Plan (VA20)	Brain MR	see [4]
[62]	CFR	CF Summary-Formative (VA20)	Brain MR	see [4]
[63]	CFR	CF Summary-Summative (VA20)	Brain MR	see [4]

13.8 Documentation of Compliance with Standards and Common Specifications

[64]	CR	Compliance Report IEC 62304	Brain MR	see [4]
[65]	CR	Compliance Report IEC 82304-1	Brain MR	see [4]
[66]	CR	Compliance Report IEC 62366-1	Brain MR	see [4]
[67]	CS	DICOM Conformance Statement	AIRC	see [4]

13.9 Clinical Data

[68]	CEP	Clinical Evaluation Plan		Brain MR	see [4]
[69]	CER	Clinical Evaluation Report		Brain MR	see [4]
[70]	SE	Scientific Evaluation (Brain-Morpho)	VA2x	Brain MR	see [4]
[71]	SE	Scientific Evaluation (WMH)	VA40	Brain MR	see [4]

13.10 Design Transfer

[72]	TR	Test Record - Design Transfer	Brain MR	see [4]

13.11 Post Market Surveillance

Plans:

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[73]	PMSP	Post Market Surveillance Plan	Brain MR	see [4]
[74]	PMCFP	Post Market Clinical Follow-up Plan	Brain MR	see [4]

Records:

[75] PMSR Post Market Surveillance Records *

Latest available reports, at the time of revision of this document are listed below:

a) OTHER_DS-Complaints_Auth.DB.Screening_Record_DH-Products-2022_2023-05-31 (*Time period: calendar year 2022*)

b) OTHER_AI-RC_PMS_Trending_Non-serious-incidents_2023-03-31 (*Surveillance Period 2022-04 - 2023-03*)

c) OTHER_AI-RC_PMS_Trending_Complaint-Investigation_2023-03-31 (Surveillance Period 2022-04 - 2023-03)

[76] PMCFR Post Market Clinical Follow-up Report

Latest available report, at the time of revision of this document are listed below:

eDoc ID: AIRC.MR.BrainMorph_GenDOC_PMCF_Report:PMS/R2.0

Scope: Brain MR VA3x & VA4x / Released: June 19th, 2023

[77] PSUR Periodic Safety Update Report:

AIRC.MR.BrainMorph_GenDOC_PSUR:PMS/R2.0

13.12 Labels and User Documentation

[78]	PLS	Product Labeling Specification	Brain MR	see [4]
[79]	IFU	Instructions for Use (IFU- EN-ROW & other languages)	Brain MR	see [6]
[80]	IFU	Instructions for Use (IFU) Engine: AIRC.Platform.WFL_UD_IFU_EN:VA31A/R3.0		
[81]	NLS	Native Speaker eIFU review	Brain MR	see [4]
[82]	R4D	Release for Distribution (R4D) - IFU	Brain MR	see [4]
[83]	DS	Datasheet	Brain MR	see [6]
[84]	UD	Addendum Online Help	AIRC	see [6]
[85]	UD	Release Information	Brain MR	see [6]

13.13 Service Documentation

[86]	ISM	Installation and Deployment Instructions	Brain MR	see [4]

13.14 DoC and Certificates

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[87]	EU Quality Management System Certificate (MDR):		
[88]	ISO 13485:2016 Certificate:		
[89]	MDSAP ISO 13485:2016 Certificate:		
[90]	DIN EN ISO/IEC 27001:2017		
[91]	EU Declaration of Conformity (DoC)	Brain MR	see [4]
[92]	Declaration of Conformity with Standards (DoCwS)	Brain MR	see [4]
[93]	General Safety and Performance Requirements Checklist (GSPR)	Brain MR	see [4]

13.15 Others

[94]	MRR	Main Review Record (R5)	Brain MR	see [4]
[95]		eIFU Compliance Report – (EU) 2021/2226	AIRC	see [4]
[96]		eIFU Risk Assessment	AIRC	see [95]

13.16 Design changes

[97]	SoU	Scope of Update	Brain MR	see [4]

13.17 Procedures and Processes

[98]	DH QR 04	Design Control
[99]	DH PLM D 124	Software Development
[100]	DH QR 05	Document and Review Management
[101]	DH QR 08	Identification and Traceability
[102]	DH QR 15	Changes Management
[103]	DH QR 16	Product Risk Management
[104]	DH PLM D 102	Configuration Management
[105]	DH QR 29	Operations
[106]	DH PLM D 133	Digital Delivery
[107]	DH QR 31	Post-Market Surveillance
[108]	SHS QT QR 25	Product and Solutions Security (PSS)
[109]	SHS QT QR 25 A2	Security Vulnerability Evaluation Guidance
[110]		DH QMS Glossary

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14 Revision History

Revision	Author(s) Changed by	Change(s)
1.0	Lesly Ipsen	First version of document released in scope of technical file review for Legal Manufacturer Transfer (LMT) project. Updated document, based on STED for AI-Rad Companion Brain MR VA40x, R3.0. Following are the only modifications introduced (marked in yellow within the document): §2.1.2. Product identification: new Basic-UDI, GTIN / UDI-DI added. Added MDT code 2010. §3.1. Example of label updated in Figure 2. §4.4. Updating manufacturer name and address, as well as facility name. §10. New SRN added, due to new manufacturer registration. §13.15. Adding reference to eIFU related documents (that were additionally provided during former technical file review for VA40 version).

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